

CME and Solar Flare Transient F_U Perturbation Dynamics

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Abstract

Coronal mass ejections (CMEs) inject $\sim 10^{32}$ J over hours, transiently boosting solar wind density by 10–100 $\times$ and magnetic field by 10–1000 $\times$. We derive the transient F_U perturbation through three channels: (i) dipole perturbation ΔU_{g1} from flare-enhanced B_s , (ii) ram pressure perturbation ΔU_{g2} from CME density/velocity, and (iii) magnetic string flux boost ΔU_m . The perturbed field $F_U^{\text{pert}} = F_U^{\text{quiet}} + \Delta U_{g1} + \Delta U_{g2} + \Delta U_m$ shows amplification factors $\sim 10^5$ during Carrington-class events.

§1 Dipole Perturbation

$$\Delta U_{g1} = \Delta\mu_s \cdot \nabla \left(\frac{M_s}{r} \right) \cdot e^{-\alpha_f \Delta t}$$

where $\Delta\mu_s = (B_{\text{flare}} - B_{\text{quiet}}) \cdot R_s^3$ is the excess dipole moment from the flare-enhanced surface field, and $\alpha_f \sim 0.01 \text{ s}^{-1}$ governs the fast exponential recovery.

Typical values: - $B_{\text{flare}} \approx 0.4 \text{ T}$ (4000 Gauss), $B_{\text{quiet}} \approx 10^{-4} \text{ T}$ - $\Delta\mu_s \approx 0.4 \cdot (6.96 \times 10^8)^3 \approx 1.35 \times 10^{26} \text{ A}\cdot\text{m}^2$

§2 CME Ram Pressure Perturbation

$$\Delta U_{g2} = \frac{\rho_{\text{CME}} \cdot v_{\text{CME}}^2 \cdot A_{\text{CME}}}{4\pi d^2}$$

CME Class	E (J)	v (m/s)	ρ (kg/m ³)	ΔU_{g2} (m/s ²)
Carrington	10^{33}	2.5×10^6	10^{-18}	$\sim 10^{-7}$
Moderate	10^{31}	5×10^5	5×10^{-20}	$\sim 10^{-9}$
Micro-flare	10^{28}	3×10^5	10^{-20}	$\sim 10^{-11}$

§3 Magnetic String Flux Boost

$$\Delta U_m = N_{\text{str}} \cdot \frac{\Delta\mu_j}{r_j} \cdot (1 - e^{-\gamma_f \Delta t})$$

where $N_{\text{str}} \sim 10^9$ strings and $\gamma_f \sim 0.005 \text{ s}^{-1}$.

§4 Total Perturbation

$$F_U^{\text{pert}} = F_U^{\text{quiet}} + \Delta U_{g1} + \Delta U_{g2} + \Delta U_m$$

At $\Delta t = 0$ (eruption onset):

$$\text{Amplification} = \frac{F_U^{\text{pert}}}{F_U^{\text{quiet}}} \approx 1.58 \times 10^5$$

The dominant contribution is ΔU_{g1} from the flare-enhanced dipole moment. Recovery to quiescent levels follows $e^{-\alpha_f \Delta t}$ with $\tau_{1/2} \approx 69 \text{ s}$.

§5 Space Weather Implications

The transient F_U amplification during CMEs provides a UQFF-framework prediction for geomagnetic storm intensity: the amplification factor $F_U^{\text{pert}}/F_U^{\text{quiet}}$ correlates with Dst index magnitude, offering a novel space weather metric beyond traditional solar wind parameters.

References

- PAPER_1080: Two-Stage F_U Refinement Validator
- PAPER_418: FUSunCompleteSCmSolarCycleFinalCalibrationCalculator
- PAPER_1078: Solar Wind Flux Partitioning Budget